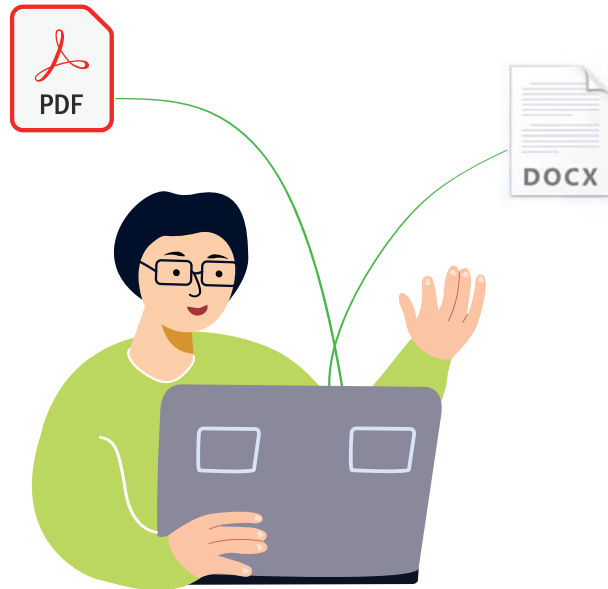


How to Create PDFs and DOCX Files for Free

Free and open source software can help you automate document creation in the most popular formats.



Governments, businesses and other organisations create a lot of documents. When they automate their systems, they purchase software components that cost hundreds or even thousands of dollars per year or per computer. It seems like a criminal waste.

I have self-published over two dozen books. They were all created using only free and open source software (FOSS). I find that other self-published authors have spent thousands of dollars to publish just one book. This does not include marketing. I am horrified when I read or hear about their expenses.

Why not pay some attention to free choices before splurging so much money on proprietary software? In this article, I will describe how anyone can create PDF and DOCX files for free. The entire toolchain will be FOSS. This process can be easily automated and integrated with application servers.

Use CommonMark (Markdown) for the source document

You can write your source documents in plain text, markup, markdown or directly in rich text. HTML (HyperText Markup Language) is markup. It is displayed as rich text. Rich text is what you see in a browser or document editor such as LibreOffice Writer. Markup is what you see when you do *View » Source* on a web page. The ODT file that you create in Writer is actually a renamed zip file with mostly XML files (eXtended Markup Language).

Markdown is the opposite of markup. It uses ordinary plain-text enhancements to unobtrusively mark up text. Before the World Wide Web came into existence, the internet was mostly email. Email, BBS and Usenet (newsgroups) users developed a plain-text style of formatting for their messages. For example, bold type was wrapped in ***asterisks*** and italicised text was wrapped in underscores. When John Gruber and Aaron Swartz created Markdown in 2004, they extended this style even further. It is now the most popular form of markdown. Markdown was released as a Perl script. Markdown documents were usually saved with the extension *.md*. I wrote my first book in Markdown and converted it like this:

```
perl markdown.pl jokebook.md > jokebook.html
```

Other implementations of Markdown were based on the Perl script but there were some differences. Finally, in 2019, Jeff Atwood and John MacFarlane published a standardised implementation of Markdown called CommonMark. I learned about it sometime in 2020 and earned the bragging rights for the first book on CommonMark. It is called ‘CommonMark Ready Reference’. Its digital version is available for free on almost all ebook stores (except Kindle). However, I will give you a quick intro so that you have an idea what it looks like.